

Misr University for Science and Technology



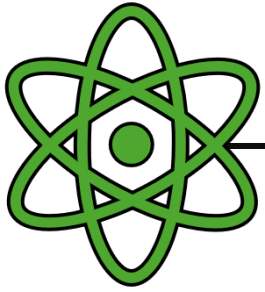
“Information Technology”

(Phy_101)

Presented by:

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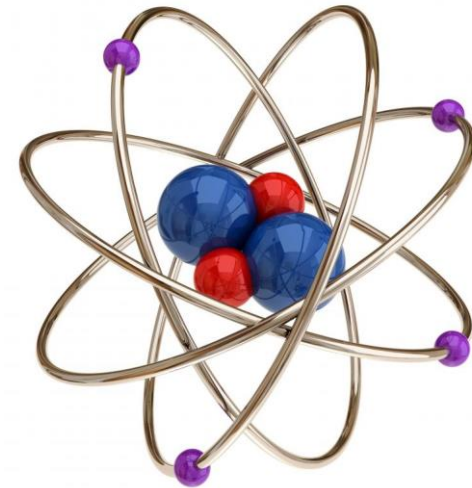


Lecture 1: Electric Forces (Part 1)



Outlines:

- ☐ *Electric Charges.*
- ☐ *Types of materials.*
- ☐ *Law of Conservation of Charge.*
- ☐ *Methods of charging.*
- ☐ *Coulomb's law.*
- ☐ *Vectors form of Coulomb's law.*



What is Matter?

❑ **Matter** is a *anything* that occupy *space* and has *mass*.

“It can be *solid, liquid or gas* in state”

○ *All forms* of *matter* are made up of *molecules*.

○ In turn these *molecules* are made up of *atoms*.

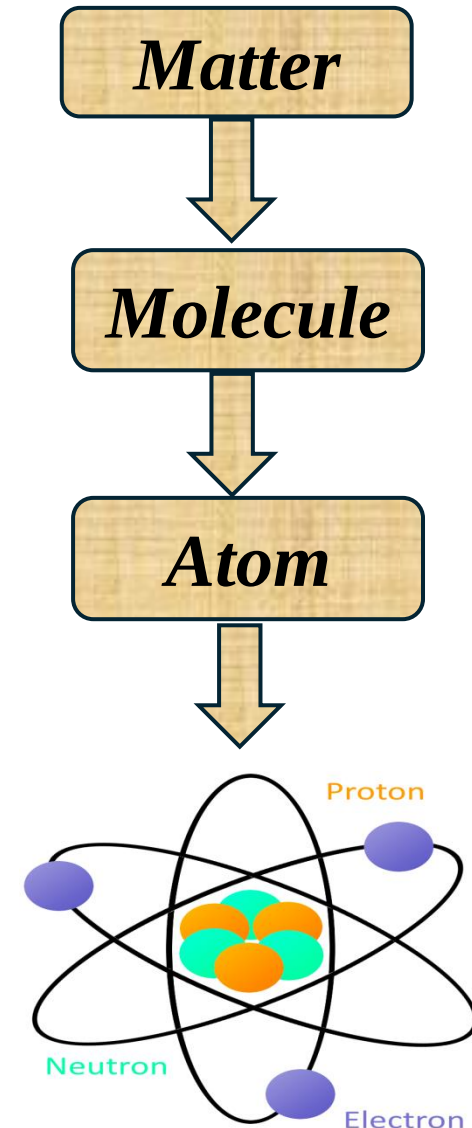
❖ *Atoms are made up:*

○ *Protons*

○ *Neutrons*

○ *Electrons*

“*Each atom has the same number of protons and electrons*”



Proton and Electron?

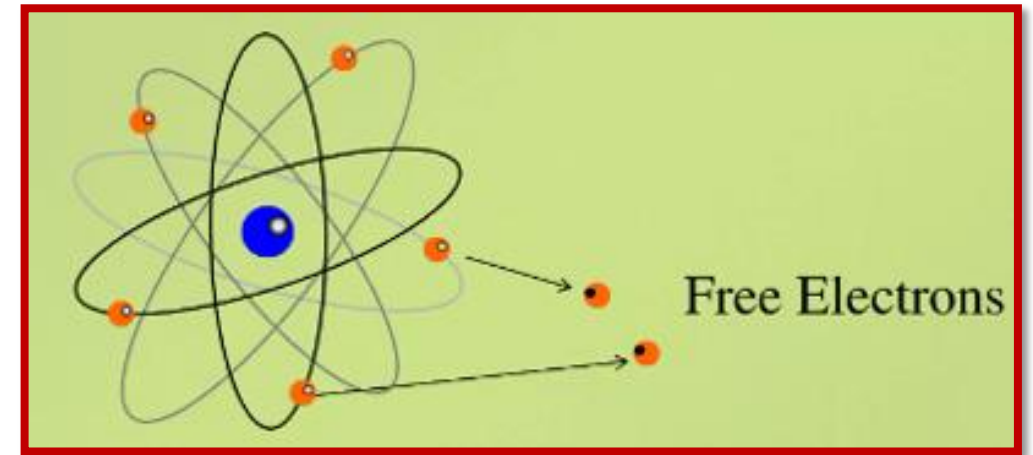
❑ Proton and Electron:

❖ **Protons** carries **positive** charge

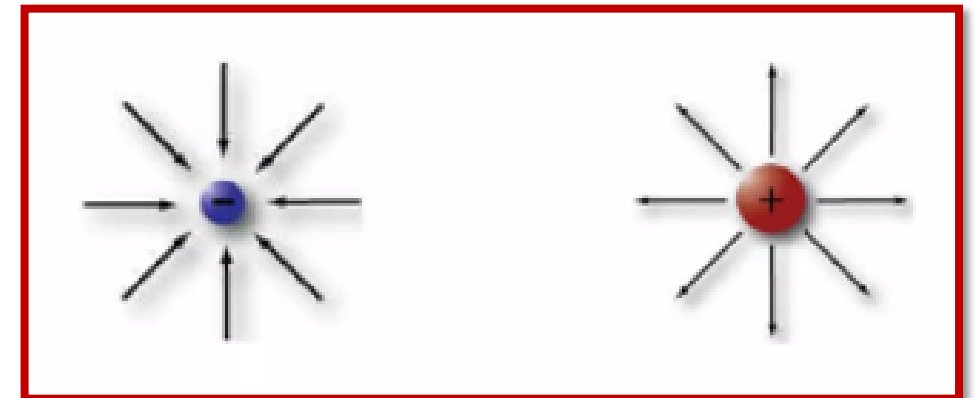
- It is relatively **large mass**.
- Does not play active part in **electrical** current flow.

❖ **Electrons** carries **negative** charge

- **Light mass**.
 - Play an important role in **electrical** current flow.
- ### ❖ **Neutral (No_charge)**



Applying Heat or Light



Charges repel and attract ?

□ Charge:

- Most **basic quantity** in an **electrical property**, measured in **coulombs (C)**.

❖ Like charges repel, unlike charges attract

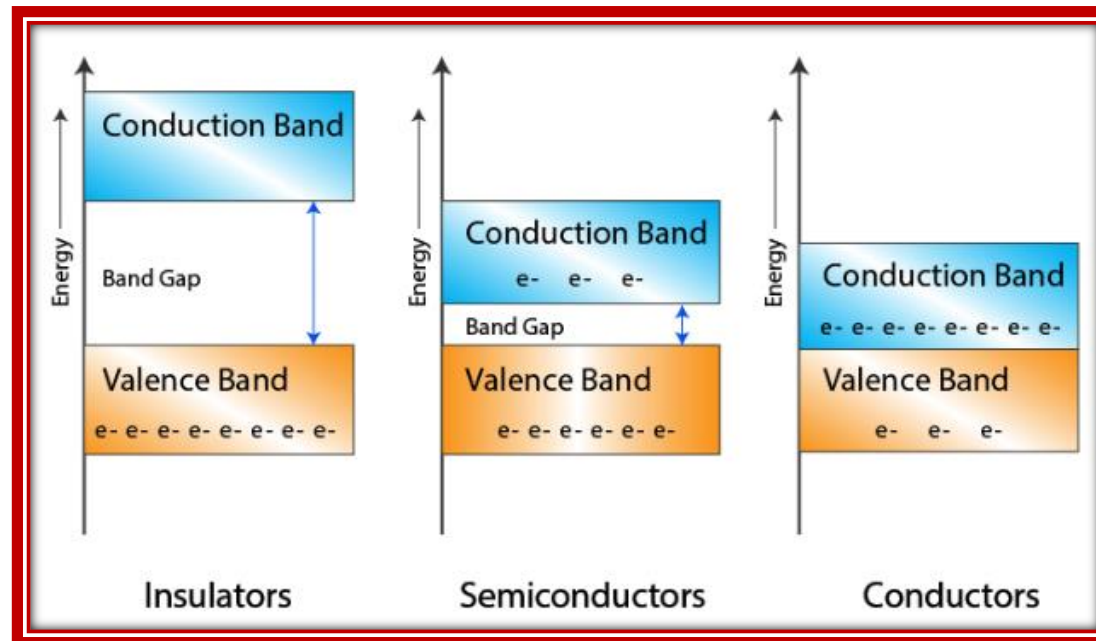
- **Electrons and protons** attract each other **because of** their **unlike** charges.

<i>Particle</i>	<i>Charge (C)</i>	<i>Mass (kg)</i>
<i>Electron (e)</i>	-1.60×10^{-19}	9.10×10^{-31}
<i>Proton (P)</i>	1.60×10^{-19}	1.672×10^{-27}
<i>Neutron (n)</i>	0	1.674×10^{-27}

What is Electrical Materials?

❑ *All materials may be classified into **three** major classes:*

Conductors	Insulators	Semiconductors
○ Conduct electricity	○ Obstruct electricity.	○ Falls between conductor and insulator
○ Have free electrons	○ Not free electrons.	
○ Example: Copper, Silver.	○ Example: Glass, Plastic	○ Example: Germanium, Silicon



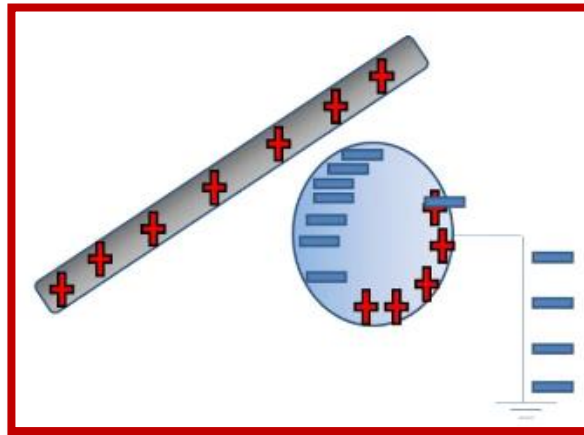
Charging Process

Methods of Charging

1. Charging by Friction



2. Charging by Induction



3. Charging by conduction

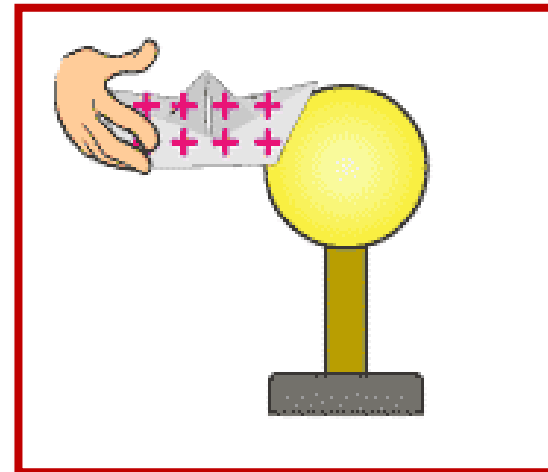


Table 1 Electrostatic Series

Material	Charge tendency
human skin	+ (weaker tendency to gain electrons)
rabbit fur	
acetate	
glass	
human hair	
nylon	
wool	
cat fur	
silk	
paper	
cotton	(stronger tendency to gain electrons)
wood	
amber	
rubber balloon	
vinyl	
polyester	
ebonite	-

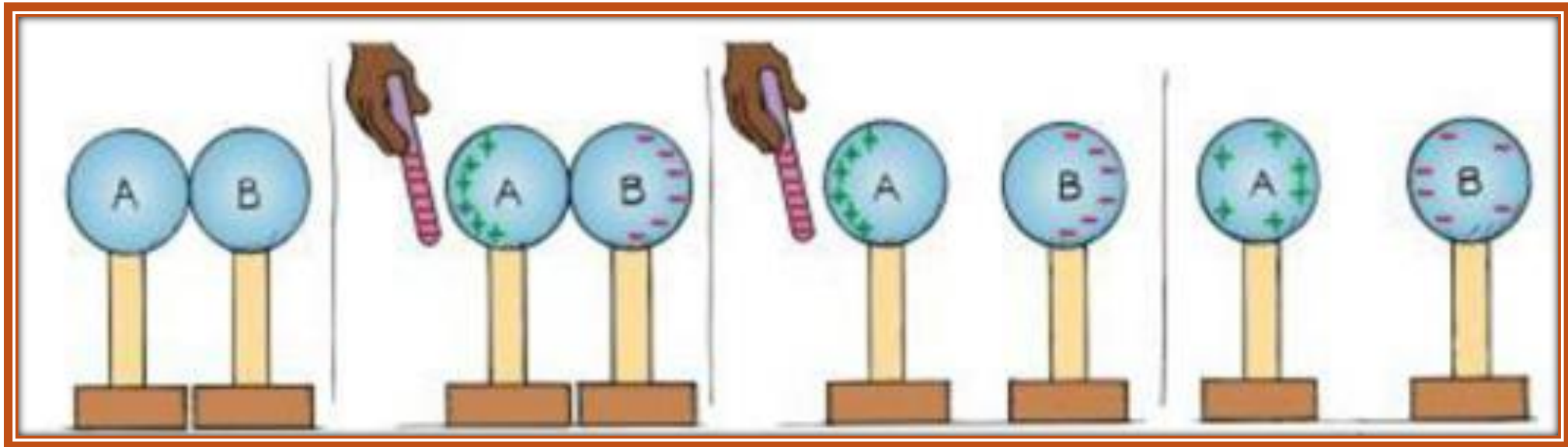
Charging by Friction

- ❑ *Transfer of electrons by **friction** between two objects.*
- ❑ *Electrons move from **your hair** to **the balloon** because the **atoms** in rubber have a **greater attraction** for electrons than atoms in hair.*
- ❑ *The **balloon** picks up a net negative charge. **Your hair** loses electrons; it becomes positively charged.*



Charging by Induction

- ❑ **Induction:** is charging *of an object without direct contact*.
- ❑ “If a charged **object** is brought *near a* conducting surface , *even* without **physical contact**, electrons *will move in the* conducting surface”.



Charging by conduction

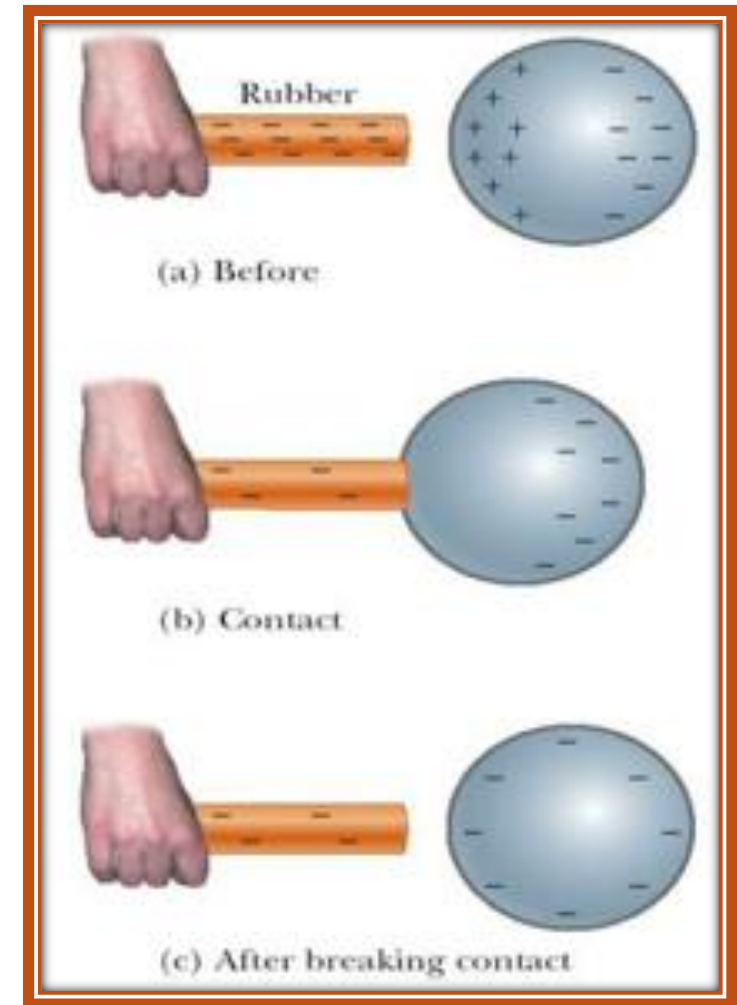
❑ **Conduction**: is the process by which charge is **transferred** from one **object to another**.

a. A charge object (the rod) is placed in **contact with** another object (the sphere)

b. Some **electrons** on the rod can **move** to the **sphere**.

c. When the rod is **removed**, the **sphere** is left with a **charge**.

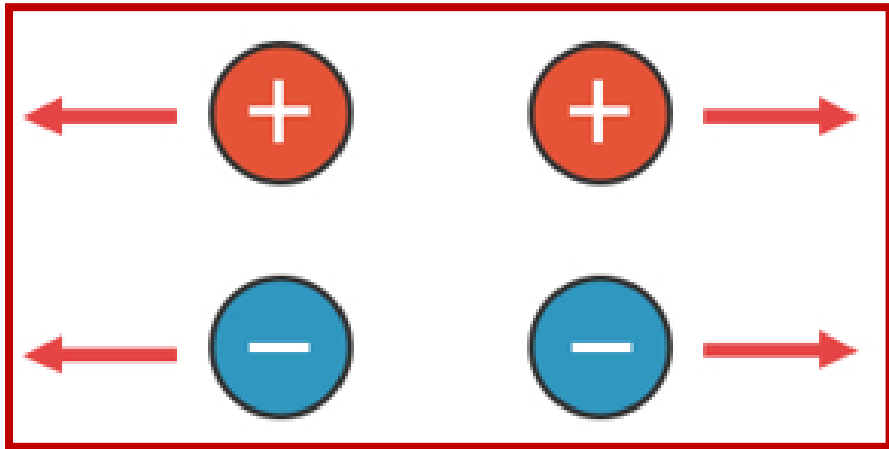
“The object being charged is always left with a charge having the same sign as the object doing the charging”.



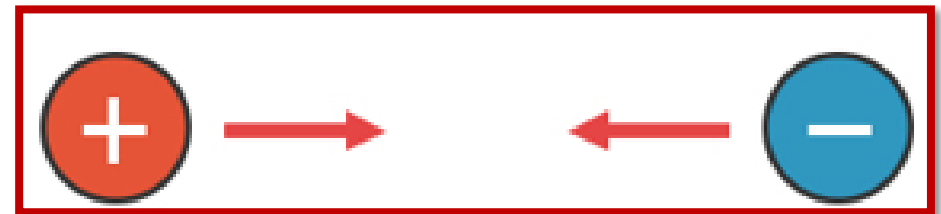
Electric Force

- *The two fundamental charge interactions are:*
 - *Attraction Force*
 - *Repulsion Force*
- *These interactions resulted in an electrical force between the two charged objects.*

Repulsion Force



Attraction Force



Coulomb's Law

□ Coulomb's Law: gives the *electric force* between two-point charges.

$$F = k \frac{q_1 q_2}{r^2}$$

$$k = \frac{1}{4\pi\epsilon_0}$$

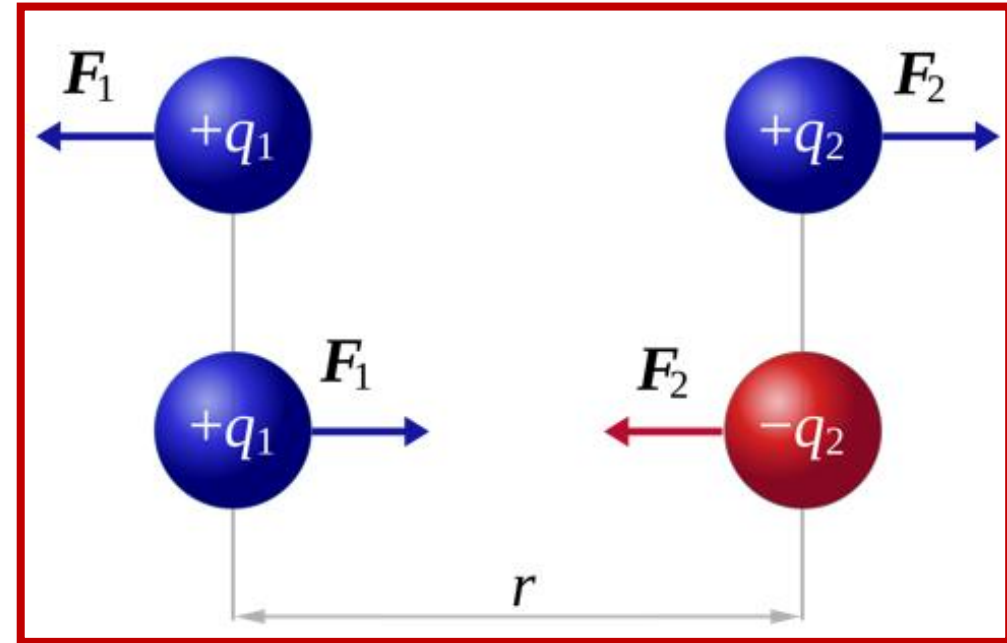
Where,

k = Coulomb's constant = $9 \times 10^9 \text{ (N.m}^2/\text{C}^2)$

q_1 = charge on mass 1

q_2 = charge on mass 2

r = the distance between the two charges



ϵ_0 is the permittivity of free space

$$= 8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$$

Factors affecting the electrostatic force

$$F = k \frac{q_1 q_2}{r^2}$$

- ❑ *The force* is *inversely* proportional to *the square* of the separation *r* between the charges and directed along the line joining them. “*It obeys the “inverse square law”*”
- ❑ *The force* is *directly* proportional to the product of the charges, *q1* and *q2*, on the two particles.

Factors affecting the electrostatic force

□ **Force** is **vector** quantity (has **magnitude** and **direction**)

$$F = \sqrt{F_x^2 + F_y^2}$$

..... (**magnitude**)

$$\theta = \tan^{-1} \left(\frac{F_y}{F_x} \right)$$

..... (**Direction**)

electrostatic force and the gravitational force

$$F = G \frac{m_1 m_2}{r^2}$$

Where,

F is ONLY an attractive force

G small magnitude

$$F = k \frac{q_1 q_2}{r^2}$$

F is an attractive or repulsion force

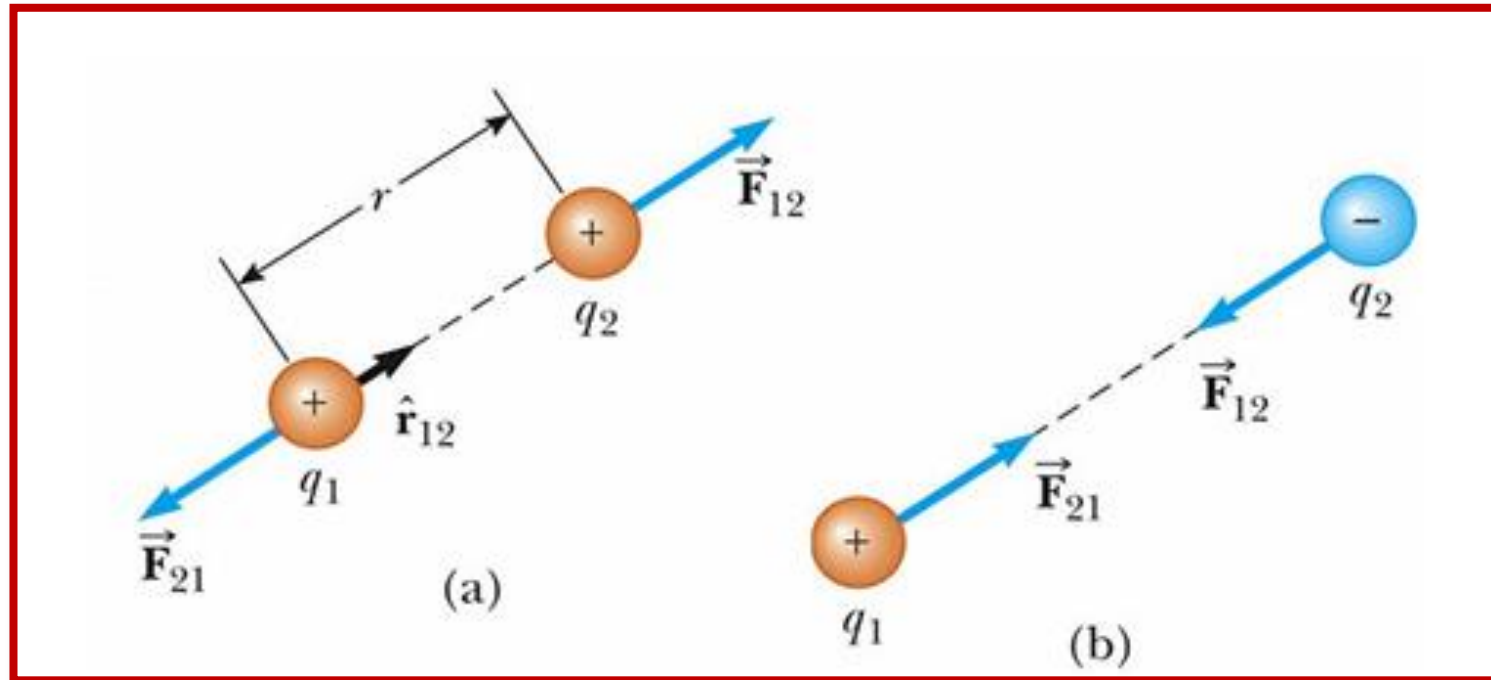
k large magnitude

- *The electrical force is much stronger than the gravitational force.*

Vectors form of Coulomb's Law

$$\vec{F}_{12} = k_e \frac{|q_1||q_2|}{r^2} \hat{r}_{12}$$

Where, $\hat{r}_{12}$ is a unit vector directed from q_1 toward q_2 .



(a) When the charges are of **the same sign**, the force is **repulsive**.

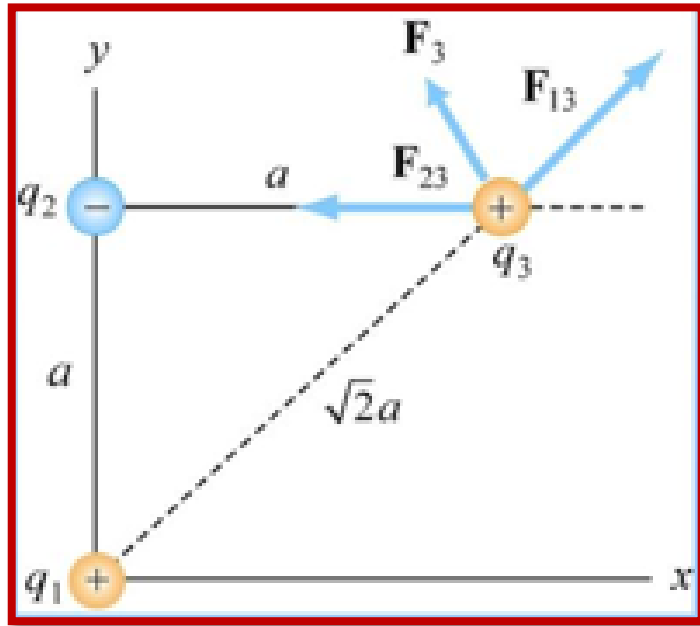
(b) When the charges are of **opposite signs**, the force is **attractive**.

Superposition principle for Coulomb's Law

❑ *Electric Forces follow the law of superposition.*

Many charges Present:

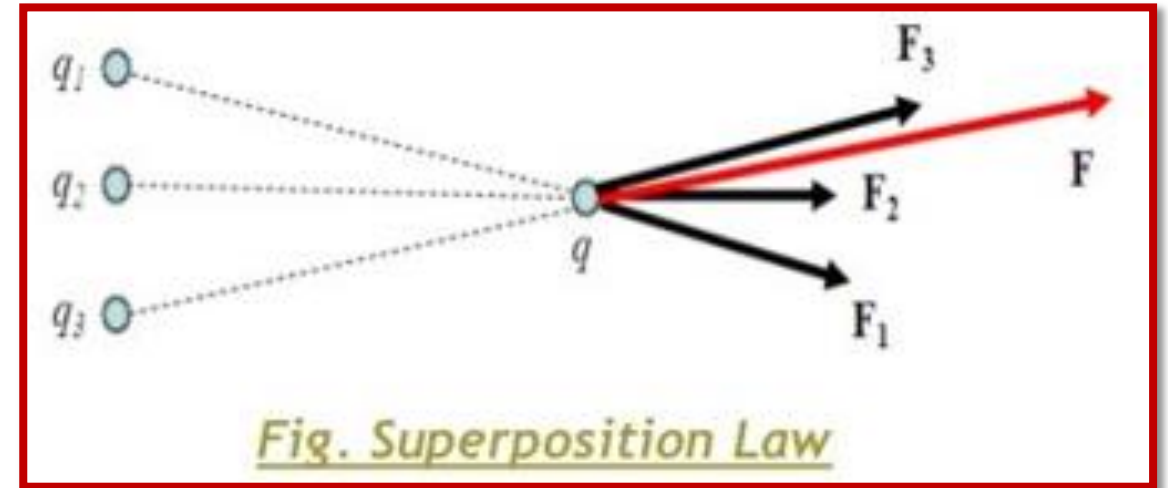
○ *Net force* on any charge is **vector sum** of forces from other individual charges.



$$\vec{F}_3 = \vec{F}_{13} + \vec{F}_{23}$$

In general:

$$\vec{F}_j = \sum_{i=1}^N \vec{F}_{ij}$$



Net force on q : $F = F_1 + F_2 + F_3$

Example 1:

□ **The electron and proton** of a hydrogen atom are separated (on the average) by **a distance $5.3 \times 10^{-11} \text{ m}$** . Find the magnitudes of the **electric force and the gravitational force** between the two particles. ($e = -1.6 \times 10^{-19} \text{ C}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$, $m_p = 1.67 \times 10^{-27} \text{ kg}$, $k = 9 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$, $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$)

Solution:

the electric force:

$$|q_1| = |q_2| = e$$

$$F_e = k_e \frac{|e||-e|}{r^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \frac{(1.6 \times 10^{-19} \text{ C})^2}{(5.3 \times 10^{-11} \text{ m})^2}$$
$$= 8.2 \times 10^{-8} \text{ N}.$$

the gravitational force:

$$F_g = G \frac{m_e m_p}{r^2}$$

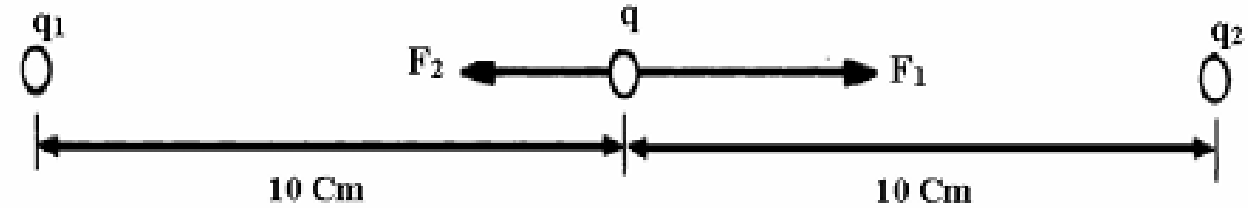
$$= (6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2) \frac{(9.11 \times 10^{-31} \text{ kg})(1.67 \times 10^{-27} \text{ kg})}{(5.3 \times 10^{-11} \text{ m})^2}$$
$$= 3.6 \times 10^{-47} \text{ N}$$

$$\text{The ratio } F_e / F_g = 2 \times 10^{39}$$

Example 2:

□ **A test charge** of $+1 \times 10^{-6} \text{ C}$ is placed halfway between a charge of $+5 \times 10^{-6}$ and a charge of $+3 \times 10^{-6} \text{ C}$ that are **20 cm** apart as in fig. **Find** the magnitude and direction of the force on the test charge.

Solution:



○ **The force exerted on the charge q_1 :**

$$F_1 = k_e \frac{|q||q_1|}{r_1^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \frac{(1 \times 10^{-6} \text{ C})(5 \times 10^{-6} \text{ C})}{(0.1 \text{ m})^2} = 4.5 \text{ N } (+i)$$

○ **The force exerted the charge q_2 :**

$$F_2 = k_e \frac{|q||q_2|}{r_2^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \frac{(1 \times 10^{-6} \text{ C})(3 \times 10^{-6} \text{ C})}{(0.1 \text{ m})^2} = 2.7 \text{ N } (-i)$$

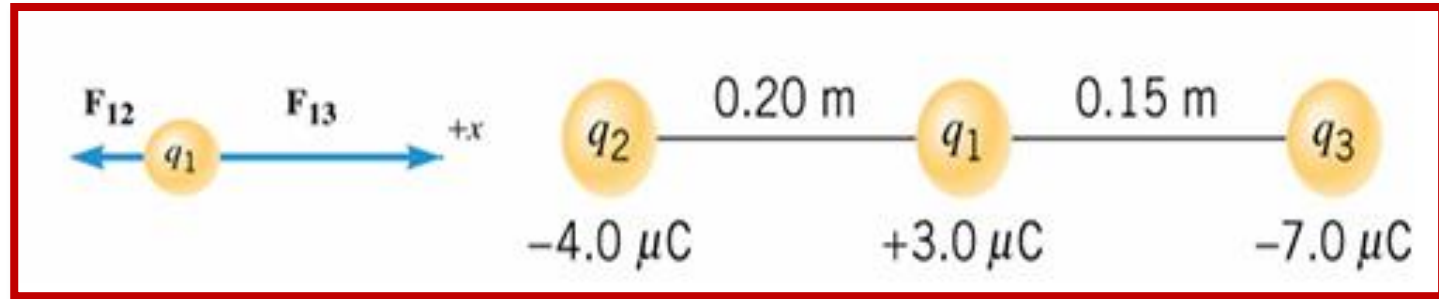
○ **The net force:**

$$F = F_1 + F_2 = +4.5 \text{ N} - 2.7 \text{ N} = +1.8 \text{ N } (i)$$

Example 3:

□ Find the total electric force on q 1 ?

Solution:



$$|\vec{F}_{12}| = k \frac{|q_1||q_2|}{r^2} = \frac{(9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(3.0 \times 10^{-6} \text{ C})(4.0 \times 10^{-6} \text{ C})}{(0.20 \text{ m})^2} = 2.7 \text{ N}$$

$$\vec{F}_{12} = 2.7 \text{ N } (-\hat{i})$$

$$|\vec{F}_{13}| = k \frac{|q_1||q_3|}{r^2} = \frac{(9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(3.0 \times 10^{-6} \text{ C})(7.0 \times 10^{-6} \text{ C})}{(0.15 \text{ m})^2} = 8.4 \text{ N}$$

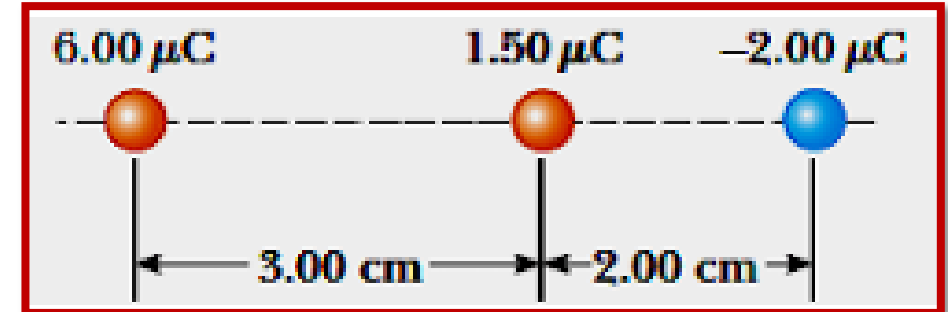
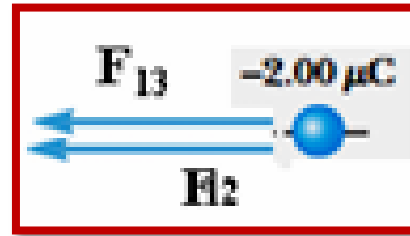
$$\vec{F}_{13} = 8.4 \text{ N } (\hat{i})$$

$$\sum F = -2.7 \text{ N} + 8.4 \text{ N} = 5.7 \text{ N}$$

Example 4:

□ Determine the electric force on the charge $-2.00 \mu\text{C}$?

Solution:



$$|\vec{F}_{12}| = k \frac{|q_1||q_2|}{r^2} = \frac{(9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(6.0 \times 10^{-6} \text{ C})(2.0 \times 10^{-6} \text{ C})}{(0.05 \text{ m})^2} = 43.2 \text{ N}$$

$$\vec{F}_{12} = 43.2 \text{ N } (-\hat{i})$$

$$|\vec{F}_{13}| = k \frac{|q_1||q_3|}{r^2} = \frac{(9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.5 \times 10^{-6} \text{ C})(2.0 \times 10^{-6} \text{ C})}{(0.02 \text{ m})^2} = 67.5 \text{ N}$$

$$\vec{F}_{13} = 67.5 \text{ N } (-\hat{i})$$

$$\sum F = -43.2 \text{ N} - 67.5 \text{ N} = 110.7 \text{ N } (-\hat{i})$$

Example 5:

□ Find the total force exerted on Q3 in the next figure?

Solution:

$$F_{32} = k_e \frac{|q_2||q_3|}{0.3^2}$$

$$F_{32} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \frac{(50 \times 10^{-6} \text{ C})(65 \times 10^{-6} \text{ C})}{(0.3 \text{ m})^2} = 325 \text{ N}$$

The magnitude of F_{31} is

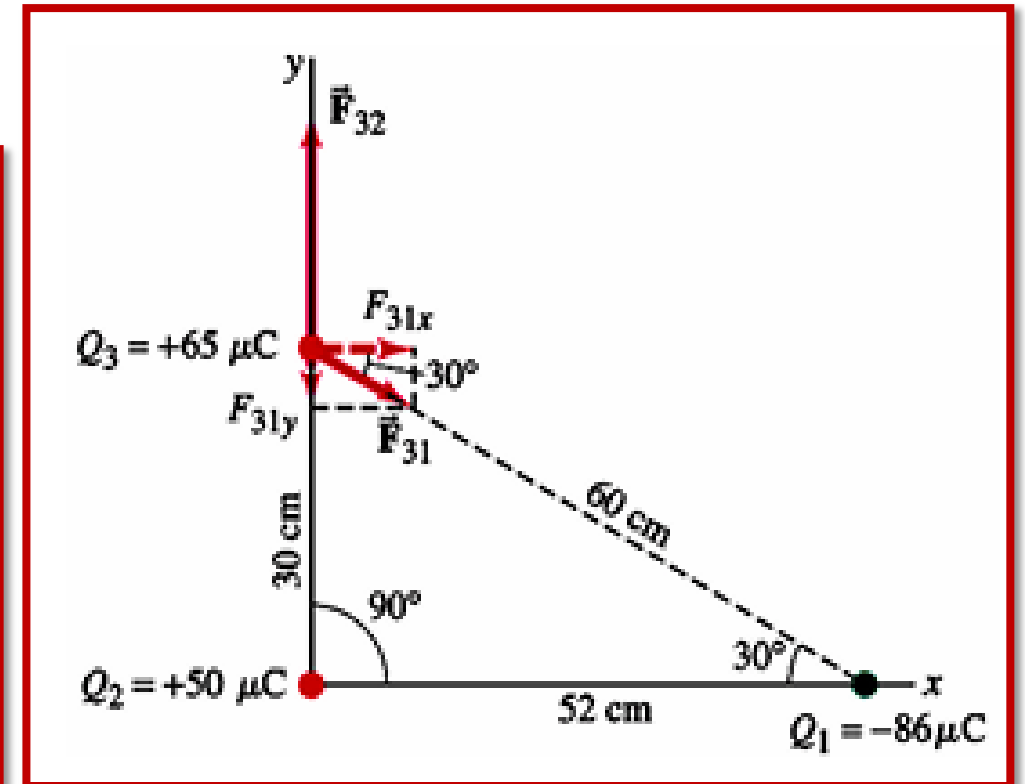
$$F_{31} = k_e \frac{|q_1||q_3|}{(0.6)^2}$$

$$= (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \frac{(86.0 \times 10^{-6} \text{ C})(65.0 \times 10^{-6} \text{ C})}{(0.60 \text{ m})^2} = 139.75 \text{ N}$$

$$F_{1x} = F_1 \cos 30 = 121 \text{ N}$$

$$F_{1y} = F_1 \sin 30 = 69.875 \text{ N}$$

$$F_3 = (121.0 \text{ i} + 255 \text{ j}) \text{ N}$$



Example 6:

□ **Two identical small charged** spheres, each having a mass of $3.0 \times 10^{-2} \text{ kg}$, hang in equilibrium as shown in Figure a. The length of each string is 0.15 m , and the angle θ is 5.0° . Find the magnitude of the charge on each sphere.

Solution:

- **The sphere is in equilibrium, the forces in the horizontal and vertical directions equal to zero.**

$$\sum F_x = T \sin \theta - F_e = 0 \dots \dots \dots (1)$$

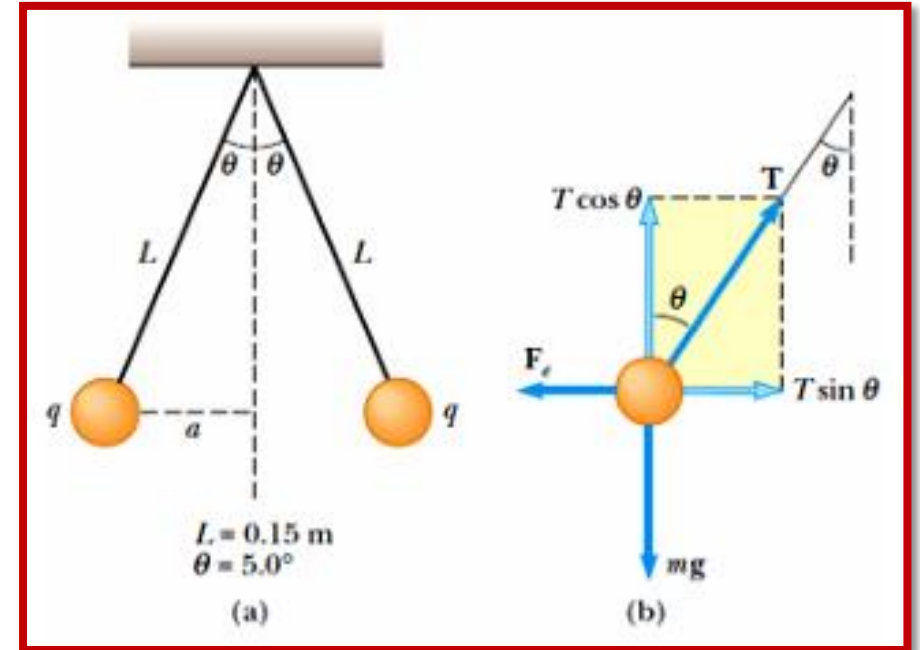
$$\sum F_y = T \cos \theta - mg = 0 \dots \dots \dots (2)$$

From Equation (2):

$$T = mg / \cos \theta$$

From Equation (1):

$$F_e = mg \tan \theta = (3 \times 10^{-2} \text{ kg})(9.8 \text{ m/s}^2) \tan(5) = 2.6 \times 10^{-2} \text{ N}$$



Example 6:

- *The geometry of the right triangle in Fig a.*

$$\sin\theta = a/L$$

$$a = L \sin \theta = (0.15) \sin(5) = 0.013 \text{ m}$$

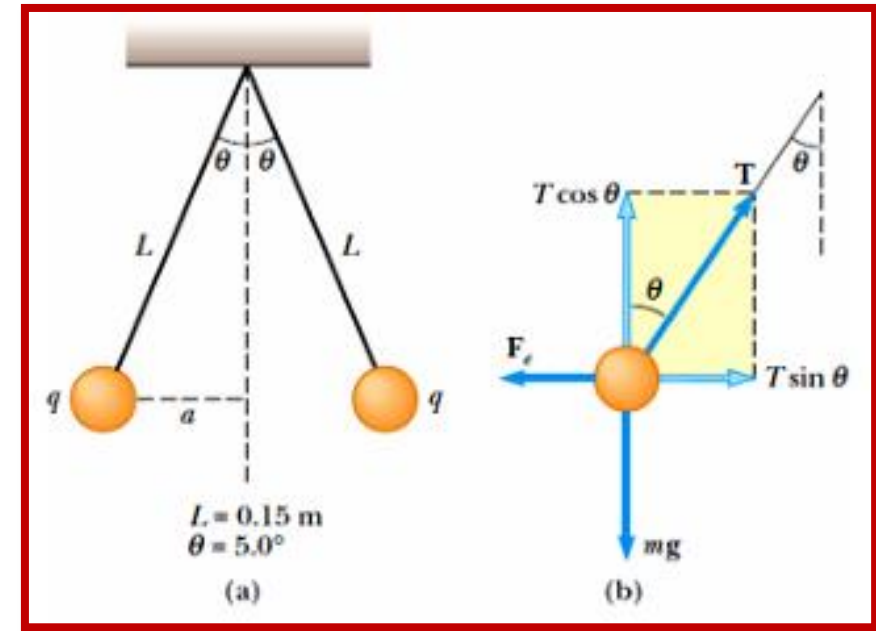
- *The separation of the spheres is*

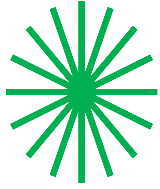
$$2a = 0.026 \quad \text{m}$$

- *From Coulomb's law, the magnitude of the electric force is*

$$F_e = k_e \frac{|q^2|}{r^2} \quad \text{And,} \quad r = 2a = 0.026 \quad \text{m}$$

$$|q^2| = \frac{F_e r^2}{k_e} = \frac{(2.6 \times 10^{-2})(0.026^2)}{8.99 \times 10^{-15}} = 1.96 \times 10^{-15} \text{ C}^2 \quad \text{And,} \quad |q| = 4.4 \times 10^{-8} \text{ C}$$





“Please don’t feel a fraid to ask”

Ref.: College Physics (Serway, Faughn).



THANK YOU ALL FOR YOUR KIND ATTENTION